

Chapter 56 — Hypospadias

- **Distal hypospadias is corrected successfully in about 80%**; the specialty's effort must now go into **reducing complications in proximal hypospadias**, where each additional procedure adds further risk.
- **Longer urethroplasty = more demanding**: a long neourethra of uniform calibre is hard to build, and the neourethra may be **inherently narrowed and unable to expand** to propel the urine bolus.
- **Poiseuille's law**: resistance to flow is **proportional to tube length** and **inversely proportional to the radius to the fourth power**. Small changes in calibre therefore have a large effect, and **the longer the tube, the greater the stricture risk**.
- Failure of the neourethra to expand → **raised resistance** → **fistula and/or diverticulum**.
- **The glans is non-distensible** (dense spongiosum). Even with uniform calibre, the **fixed distal urethra** causes turbulence and **high proximal pressure** → **diverticulum**.
- The neourethra is a **skin-lined tube without spongiosum**, an independent risk for progressive distension.
- Tissue **dysplasia**, the **severity of curvature** and **skin shortage** may impair healing, perfusion and growth with age. Each surgeon's numbers are falling, so **multi-institutional collaboration** is essential.

	Tubularised incised plate (TIP), long penoscrotal neourethra	Onlay island flap (OIF), penoscrotal neourethra
Tissue	More rigid neourethra	More pliable, elastic preputial tissue
Resistance	Fixed resistance along a long length	Lower : distends with micturition
Uroflow	Prolonged, flattened void ; risk of proximal fistula	More normal flow pattern

Figure legend, Fig. 56.32 (Braga et al., 2007) — A short coronal TIP neourethra gives a near-normal flow curve, because there is little resistance over a short distance. The meatus is assumed patent in all cases.

Figure legend, Fig. 56.30 (Jayanthi, 2017): *modified Ulaanbaatar staged repair* — First stage: correct curvature with **small intestinal submucosa** → harvest a **transverse island Duckett tube** for the glanular and distal penile urethra → tunnel it into the glans, or perform glansplasty → rotate **Byars flaps** ventrally to bridge to the native urethra → drip stent. Second stage **at 6 months**: elliptical incision around the urethral defect and tubularise. (The example case had 46,XY ovotesticular disorder of sex development.)

Figure legend, Fig. 56.31: *single-stage buccal inlay* — When the plate is **too narrow to tubularise**: U-shaped Thiersch-Duplay (TD) incision → **midline incision in the plate** → **buccal graft secured into the midline** → urethroplasty to the width of the U-incision → barrier layers → closure.

KEY POINTS

- **Proximal hypospadias has a higher complication rate** than distal.
- **The trend is towards staged repair** for severe hypospadias: **curvature is corrected at stage one, urethroplasty at stage two**. This gives supple ventral tissue for tubularisation and avoids a urethroplasty laid over a corporal graft in one stage.
- **25–75% need additional procedures** after a planned two-stage repair.

Concept — Proximal repairs fail for **physical reasons**: a long, rigid, spongiosum-free tube ending in a fixed glans. Length and stiffness raise resistance, and raised resistance produces fistulae, diverticula and strictures.

Postoperative Management

- A careless postoperative course can ruin a good repair, yet **quality evidence for specific regimens is lacking**.
- **First 48 hours**: urinary diversion plus a **compressive dressing**, which reduces stress on the neourethra, tamponades bleeding and limits swelling. The authors leave the dressing **72 hours**, then apply petroleum jelly.
- **A soft, pliable urethral stent in complicated repairs reduces meatal stenosis and fistula**. A **suprapubic tube** is used in older children and redo cases.
- **Diversion lasts 3–21 days**, by complexity, urethroplasty length and preference. Longer stenting reduced complications in one study but is not universally accepted, and must be weighed against **discomfort and UTI risk, which rises with each day**. Short distal repairs have been done **without a stent** with no change in outcome.
- **Bladder spasm** → high-pressure voiding around the stent → stress on the repair, so give **oxybutynin until the stent is removed**.
- **Long, stiff stents** press on the ventral repair and risk **glans dehiscence**. A **Foley catheter must be fixed in the midline to the anterior abdominal wall**. Tape often slips and pulls the catheter laterally or ventrally, so the authors have **abandoned Foley catheters** because of glans dehiscence.
- **Analgesia**: paracetamol is near-universal, and about half of urologists add **non-steroidal anti-inflammatory drugs (NSAIDs)**. **Ibuprofen and ketorolac do not increase bleeding**. **Regular paracetamol plus ibuprofen** minimises opioid use, tapering after **48–72 hours**.

Complications of Hypospadias Repair

- Duckett described “hypospadiology” as a **humbling blend of art and science**. The tissues are delicate and healing depends on **fragile neovascularisation**, threatened by oedema, infection and variable healing.
- **Complications have lacked standard definitions**, which makes reporting inconsistent. The standardised set is in Table 56.1.

TABLE 56.1 COMPLICATIONS AND DEFINITIONS

Complication	Standardised definition
Urethroplasty complications (primary outcome)	
Fistula	Urethral leak anywhere below the meatus
Glans dehiscence	Complete separation of the glans wings → coronal or more proximal meatus , or wings separated by a skin bridge; objectively glans fusion <2 mm
Meatal stenosis	Obstructive symptoms (straining, prolonged voiding, UTI, retention) plus calibration <8 Fr before puberty or <12 Fr after puberty
Urethral stricture	Obstructive symptoms plus near-closure of the urethra on urethroscopy
Urethral diverticulum	Segmental sacculation during voiding
Recurrent curvature	Residual ventral, dorsal or lateral curvature >30° on erection (spontaneous or artificial)
Skin complications (secondary outcome)	
Skin surplus	Excess skin ≥2 cm in a circumcised patient with the suprapubic fat pad retracted
Penile torsion	Glanular plane >30° from the vertical ; specify direction
Non-retractable reconstructed prepuce	Meatus not visible (without lichen sclerosus); acceptable before puberty, abnormal after
Preputial dehiscence	Defect recreating an incomplete prepuce

Complication	Standardised definition
Preputial fistula	Fistula-like preputial defect not communicating with the urethra
Lichen sclerosus	White cicatrix, preferably confirmed histologically
Also listed	Skin deficiency with tethering; postpubertal phimosis; inclusion cyst

Courtesy Warren Snodgrass, MD.

- **What the table is for:** a common language, so that series can be compared. The pattern is that **urethroplasty failures are the primary outcome, skin problems secondary**. The numbers to hold are **>30° for curvature and torsion, <8 Fr / <12 Fr for stenosis and <2 mm glans fusion**. The trap: **calibration alone is not stenosis**; obstructive symptoms are also required.

Urethrocutaneous Fistula

- **The commonest complication: just under 10%** at short-term follow-up. It occurs **most often at the original meatus or the coronal margin**, from pinpoint to large.
- **Causes:** ischaemia, oedema, infection, haematoma and especially **distal obstruction** (meatal stenosis or stricture → high pressure and turbulence). **Technical causes:** overlapping suture lines, inadequately inverted epithelium, poorly absorbable sutures. Risk **rises with severity**.
- It is noticed as **multiple streams**, mostly **within 2 years**, or at toilet training.
- **Small early fistulae rarely close spontaneously. Repair at least 6–12 months** after the primary operation, once tissues soften.

Assessment under anaesthesia: **bougie à boule calibration** for patency and irregular calibre → **urethroscopy** if irregular → **“stress test”**: catheter in the urethra, compress proximally and distally, inject saline or methylene blue to reveal hidden tracts.

FISTULA REPAIR BY SITE AND SIZE

Fistula	Repair
Small, shaft, proximal to the corona	Excise the epithelialised tract → subepithelial urothelial closure → multiple offset layers
Large, or thinned tissue	Trap-door or island flap of shaft skin; second layer or vascularised flap, offset
Coronal margin	Implies distal narrowing or poor tissue → revise the urethroplasty , not simple excision

- **Recurrence about 20%** even after good repair. Reduce it with **interposed vascularised non-epithelialised layers**: redundant dartos, **scrotal tunica vaginalis flap**, scrotal dartos, a de-epithelialised flap, or the **Cecil modification** when tissue is scarce.
- **Diversion:** none for simple repairs; **7–14 days** for complex redo repairs.

Figure legend, Fig. 56.34 — In fistula repair after a two-stage proximal repair, the **distance from the glans drives decision-making**. The sequence is: mark around the tract and raise flaps for barrier cover → isolate the tract → undermine → primary closure → dartos over the repair → skin closure. There was no recurrence at 6 months.

Glans Dehiscence

- **Causes:** catheter tension, poor technique, **glans closure under tension**, vascular compromise. The meatus **retreats to a coronal or subcoronal position**, in effect a breakdown of the repair.
- **Reoperate only for voiding symptoms**. If repair is needed, an **interposition inlay graft**, as a staged step or in a single stage, avoids glans tension.

Meatal Stenosis

- Definitions vary. **Risk:** urethral closure carried **too far distally**, or a **tight glans closure**. Longer stenting may help (weak evidence).
- **The postoperative meatus should calibrate ≥ 8 Fr, increasing with age.** Clinically relevant stenosis means **stranguria, deflected stream, poor flow, retention** and a **“laser-beam” stream**.
- **Escalation:**
 - **Sound dilatation $\pm$ topical betamethasone** in some cases.
 - **Routine postoperative dilatation is unnecessary**, but early dilatation may reveal subclinical stenosis.
 - **Refractory to dilatation $\rightarrow$ dorsal midline meatotomy (rarely), otherwise redo urethroplasty.**
- **Meatal retrusion:** excessive distal tension, or after glans dehiscence. It results from poor planning: an immobile or poorly vascularised urethra retracts. **Revise if the deflected stream prevents standing voiding.**

Urethral Stricture

- Rates depend on **technique and urethroplasty length**. Sites: **native-neourethral junction**, along the neourethra (especially at the **coronal margin with a narrow plate**), or within the glans. It presents with a weak stream, stranguria, retention or UTI.
- **Causes:** technique, ischaemia, trauma, infection. **Characterise by cystoscopy** (length, calibre, site).
- **Short, mildly symptomatic strictures $\rightarrow$ dilatation or endoscopic incision, rarely durable. Internal urethrotomy** succeeds poorly and can worsen scarring. Even **strictures < 1 cm with intermittent catheterisation** seldom keep normal flow. Results may be better for short anastomotic strictures treated at 3 months, but the authors rarely see success.
- **Refractory or extensive strictures $\rightarrow$ revision urethroplasty.** Expect tissue loss, so be ready to harvest **local flap or buccal mucosa**; this is often multistage.
- **Hair-bearing skin in the urethra** after multistage or complex repairs, combined with narrowing, becomes a **nidus for stones and UTI**. Options: laser epilation; endoscopic stone removal. When **stones, hair and UTI coexist $\rightarrow$ resect and replace** with buccal graft or local skin.
- **Recurrent UTI $\rightarrow$ look for diverticulum or stricture.** An **enlarged prostatic utricle** causing stasis $\rightarrow$ **excise or fulgurate**.

Urethral Diverticulum

- It presents with a weak stream, **post-void dribbling**, UTI, rarely haematuria, **ballooning of the shaft**, or the need to **“milk” the urethra**.
- **Commoner after preputial flap, two-stage and proximal repairs:** there is **no spongiosum**, and a **rigid glanular neourethra** raises penile urethral pressure.
- **Small saccular diverticulum $\rightarrow$ excise and reduce to a uniform calibre. Extensive diverticulum (more common) $\rightarrow$ excise redundant tissue, close the urethra and add multilayer reinforcement.**
- The redundant wall is **well vascularised** and can be **reused as a flap** to repair coexisting fistula or distal stricture. **Extraurethral plication** avoids entering the urethra.

Figure legend, Fig. 56.36 — Excise the excess epithelium but **keep the redundant dartos** as an extra layer over the urethral closure.

Balanitis Xerotica Obliterans

- **Balanitis xerotica obliterans (BXO), or lichen sclerosus, is a chronic inflammatory dermatosis.** It occurs with phimosis in uncircumcised boys, with meatal stenosis, or with progressive urethral narrowing after repair, appearing **pale and white**.

- **Meatal disease only** → **topical betamethasone or clobetasol**, or **systemic tacrolimus**. These **usually fail**.
- **Definitive: resect all involved tissue**, send it for histology, and replace with **inner prepuce or buccal mucosa in a staged repair**. **BXO can invade buccal mucosa**.
- **Recurrence up to 40%** on medium-term follow-up, so **monitor closely**.

BXO sequence: urethral “cutback” to healthy tissue → topical steroids → excise persistent disease → place a **buccal graft** and confirm take without recurrence → urethral closure.

Recurrent Penile Curvature

- It damages urinary and sexual function and may **worsen at puberty** through differential dorsal–ventral corporal growth.
- **Causes: curvature underestimated or incompletely corrected, graft contraction** after corporotomy, **ventral skin traction**, or a preserved **tethering plate**.
- **Assess by complete degloving with artificial erection**. A **tourniquet can hide basal curvature**, so compress the corpora against the pubis instead.

Staged correction (counsel the family that several procedures are likely): stage 1 identifies the cause (watch for **corporal disproportion after a previous dorsal plication**). If a corporal graft is used, cover it with **healthy dartos as a bed for buccal graft** → stage 2: **ventral buccal graft** unless skin allows tension-free cover → stage 3: **Cecil modification** if shaft skin is deficient → stage 4: tension-free skin closure.

Figure legend, Fig. 56.38 — A 14-year-old after **infant single-stage repair with dorsal plication** presented with marked ventral curvature. Correcting it left a **long urethral gap**, which needed a buccal inlay and then final closure.

Skin Complications

- **Buried penis with shortening** from concealment and poor skin fixation is prevented by **degloving plus anchoring sutures from Buck's fascia to the dermis** at the penopubic and penoscrotal junction.
- **Suture sinus tracts** come from thin ventral skin, full-thickness sutures, or sutures tied too tight (ischaemia).
- **Penoscrotal transposition or webbing** after proximal repair → **scrotoplasty** to prevent concealment.

Concept — *Almost every urethroplasty complication traces back to **distal obstruction, ischaemia or tension**. That is why fistula and diverticulum repair start by excluding distal narrowing, and why every repair adds vascularised barrier layers.*

HYPOSPADIAS OUTCOMES

- The literature is **weak**: retrospective, small, technically varied, with poorly defined outcomes, no independent assessment and much loss to follow-up.
- **Largest meta-analysis of TIP repair** (Pfistermuller et al., 2015; 49 studies, 4,675 boys, mean follow-up 16 months): **complications 10.6%, reoperation 4.5%; fistula 5.7% > meatal stenosis 3.6% > stricture 1.3%**. An **extra dartos layer improved outcomes; proximal and redo cases did significantly worse**.
- **TIP vs Mathieu** (meta-analysis): **similar overall complications and fistula rates**, but **meatal stenosis much commoner after TIP**.
- **Recurrent UTI after repair is uncommon (about 2%)**. Imaging finds distal narrowing, VUR, diverticulum or an enlarged utricle. **TIP → raised post-void residual; OIF → diverticulum or fistula**. **Image anatomically any child with recurrent UTI**.

- **Distal hypospadias:** success **83–95%**. Complications in large single-centre series are higher with **longer follow-up and a more proximal meatus**. After TD repair, **fistula and glans dehiscence occur early (first year) but stricture late**.
- **Proximal hypospadias:** complications **23–68%**; **fistula is again commonest (3–45%)**, then glans dehiscence and meatal stenosis. **Diverticula are commoner after Byars flap or onlay** than after two-stage graft repair (STAG or Bracka).
- **Recurrent curvature after two-stage repair is low (0–10%)**. Two-stage repair with corporoplasty gives **greater penile length and better cosmesis**. Penile length is the chief concern of adults repaired in infancy.
- **Staged tubularised autograft (STAG):** complications **<25%**, mainly glans dehiscence or fistula; recurrent curvature low. Extra layers and better blood supply reduce complications.
- In a four-way comparison (TIP, preputial island onlay, island flap tube, two-stage Bracka), **the approach did not change complications**. **Dorsal plication plus proximal urethral mobilisation** corrects most curvature and avoids laying a skin graft over a corporal graft.
- **Byars flap two-stage repair: high complications (30–70%)**, mainly fistula then glans dehiscence. Diverticulum results from **no spongiosum plus no fixation of the flap** to the ventral shaft.
- In one centre, **single-stage repairs had more complications than planned two-stage repairs** (not significant). Two-stage approaches that close the distal urethra and correct curvature at stage one report favourable results.
- Penile perception scores were **better with staged repair**, which gave a **longer penis**, with similar complications and urinary complaints across four techniques.
- **No direct comparison** identifies the ideal approach to proximal hypospadias with severe curvature, and **long-term data are lacking**.

FIG. 56.39 SUMMARY OF RECENT STUDIES ON PROXIMAL HYPOSPADIAS OUTCOMES — REBUILT BY TECHNIQUE

Repair	Reported complication rate (follow-up about 1.5–5 years)	Note
Two-stage Byars flap	43–81% across four centres	Highest-risk staged option
Long single-stage TIP	About 60%	—
Single-stage (mixed)	About 60%	—
Dorsal inlay graft	About 30–50%	—
Pedicled preputial flap	About one-third	—
Two-stage repair (graft)	About 38%	—
Proximal TIP	About 25%	No longer recommended: higher recurrent curvature
STAG	About 23%	Lowest reported
STAC	Unpublished	The authors' current treatment of choice

Data from Stanasel et al., 2015; McNamara et al., 2015; Pippi Salle et al., 2016; Long et al., 2017; Snodgrass et al., 2017, 2021; Chan et al., 2021.

PROXIMAL HYPOSPADIAS COMPLICATION RATES: FIGURE VERSUS BODY TEXT

Series	Fig. 56.39	Body text	Reason
Boston, two-stage Byars	53% (follow-up 40 months)	49% (follow-up 46 months)	Different data cut or denominator
Toronto, long TIP / staged	61% / 38%	53% / 32% (follow-up 30–48 months)	Different denominator (140 patients in the text vs per-technique counts in the figure)

- **What the figure tests: Byars-flap two-stage repairs carry the highest complication rates, and graft-based staged repairs (STAG) the lowest.** Every proximal technique still fails in roughly a quarter to two-thirds. The trap: “complication” is often defined as **any unplanned extra procedure**, so rates are not directly comparable.

Outcome Assessment

- Fistulae are obvious; **early stricture is subtle**. Watch the stream in clinic or on **family video**. Office calibration is difficult, so **examine under anaesthesia** if concerned.
- **Uroflow** helps, but normal values after repair are poorly defined, it cannot be done before toilet training, and **flow improves with age**.
- **Only about half of complications appear in the first year**; about three-quarters are found by 3 years. **Fistulae appear early and strictures late**.
- **Authors' standard:** follow **through puberty**, especially proximal cases. Review after toilet training (stream), and at puberty (velocity, curvature, sexual function).

Patient-Reported Outcomes

- **Surgeon-judged success may differ from patient and parent perception.** Parental **decisional regret** follows complications, hesitancy about surgery and a wish for circumcision. **Better education reduces conflict**.
- **Decision aids** (an online “hypospadias hub” of video and text): families engaged repeatedly, felt confident, and had **less decisional regret**.

HYPOSPADIAS OUTCOME INSTRUMENTS

Instrument	Rated by	Distinguishing feature
HOSE (Hypospadias Objective Scoring Evaluation)	Surgeon	Objective surgical outcome score
Pediatric PPS (Penile Perception Score)	Patient and parents	Satisfaction with length, meatus, glans, skin, straightness
GPS (Genital Perception Score)	Patient	Genital self-perception
HOPE (Hypospadias Objective Penile Evaluation)	Independent panels (parents, patients, urologists) scoring standard photographs against reference pictures	High interobserver reliability

- All share components: meatus, fistula, stream, length, skin, overall appearance. A review found they **neglect sexual function and psychosocial factors** and are **biased towards surgical features** (meatal position and appearance), which may matter less to patients. Uptake of PROs is poor.

TABLE 56.2 HYPOSPADIAS OBJECTIVE PENILE EVALUATION (HOPE) SCORE

Item	10 points	Intermediate	Lowest
1. Meatal position	Position 1	Positions 2–4: 8, 5, 3	Position 5: 1
2. Meatal shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
3. Glans shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
4. Penile skin shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
5.1 Torsion	0–30°	30–50° 7; 50–70° 4	>70° 1
5.2 Curvature on erection	0–30°	30–50° 7; 50–70° 4	>70° 1 (omitted if no erection seen)

From Abbas TO, Front Pediatr 2022. HOPE score = mean points across items 1–5.

- **What it tests:** HOPE scores **appearance on a 1–10 scale per item** (higher = more normal), judged against reference photographs (Fig. 56.41). The pattern: torsion and curvature share the same **30° steps**, and **0–30° counts as normal**, matching Table 56.1's >30° complication threshold. The trap: HOPE is **cosmetic only**, with no voiding or sexual domain.
- **Hypospadias-specific health-related quality of life (HRQoL)** framework (scoping review) defines **five domains: penile appearance, sexual health, psychological and behavioural function, social function, voiding function**. These are the basis for a future patient-reported outcome measure (PROM), because current tools are insufficient (Fig. 56.42).
- **Mild, undiagnosed distal hypospadias** (no curvature) in adults: **minimal sexual, fertility or voiding problems**. Many men are unaware of it, adapt to spraying, are sexually active and father children. **More proximal defects with curvature report worse sexual and functional outcomes**.
- **After surgery:**
 - Older studies reported more behavioural and school problems (worse with severity) and later, less frequent sexual activity.
 - A contemporary controlled series found **equal fertility, sexual satisfaction and age at first intercourse overall**, but **worse satisfaction, fertility and sensation in proximal cases**.
 - Adolescents after proximal repair worry about **penile growth more than meatal position** and are more hesitant about sexual contact.
- **Overall:** no major psychopathology, but **less satisfaction with appearance**, sexual inhibition, more negative genital appraisal, anxiety and lower self-esteem. **Earlier surgery may help**. Lay observers see **distal repairs as normal** and proximal repairs as only slightly different.

KEY POINTS

- **About 60% of urethroplasty complications occur within 1 year**; stricture and curvature can appear late, so **follow proximal cases into puberty**.
- **Risk factors:** proximal meatus, glans width <14 mm, no barrier flap, previous hypospadias surgery.
- **Fistula is commonest:** primary closure if tissue is supple; **redo urethroplasty and glans closure if glans tissue is thin**.
- **Meatotomy or skin flap for stenosis usually fails with BXO**; excise all BXO and stage with buccal graft. Failure is up to 40%, so monitor.

- Long term: more **ejaculatory problems**, less sexual satisfaction, more dissatisfaction with appearance. **PROs are under-used.**

Concept — Hypospadias outcomes are **time-dependent and observer-dependent**. Complications keep appearing for years, and surgeons, patients and parents judge success differently. Only follow-up through puberty with patient-reported measures captures the true result.

Hypospadias Reoperations

- Simple problems (fistula, skin redundancy) can be fixed in one stage. **Each unplanned operation degrades tissue**, scarring the penis, disrupting the blood supply and shrinking options, so **complication rates rise with each redo**.
- **Devine (1973)**: after failure, surgeons tend to attempt “**small operations**”, which make matters worse. What is needed is “**extensive resection of scarring, major shifts of tissues and meticulous reconstruction.**”
- **Assessment**: history of voiding, skin and erectile problems; **previous operative notes** (tunica vaginalis flaps, grafts, urethroplasty type, curvature correction). Have a **low threshold for cystoscopy**, and **always perform artificial erection**, because missed curvature has severe long-term consequences.
- Blood supply of previously operated flaps is unpredictable, so **add layers (tunica vaginalis)**.
- **Tubularisation redo** (Mathieu, TD, redo TIP) has reasonable success. An **inlay graft after incising the plate** improves redo outcomes, but take care with tissue that has already failed.
- **Focal problem (fistula, diverticulum) without curvature** → local or remote skin flaps, as onlay or multilayer cover, especially from **redundant non-hair-bearing shaft skin**.
- **Split-onlay skin (SOS) flap**: a pedicled skin flap of about 1 cm, from anywhere redundant (shaft or circumcision line), **split into an onlay for the urethra plus a second piece for barrier or skin cover**, or de-epithelialised as an extra layer.

ADJUNCTS TO IMPROVE TISSUE IN REOPERATIVE REPAIR

Adjunct	Mechanism	Use and evidence
Topical nitroglycerin	Arterial and (mainly) venous dilatation → more flap blood flow (shown by fluorescein imaging)	Applied for 24 hours after repair; no randomised trial
Hyperbaric oxygen (HBOT)	Oxygen gradient drives angiogenesis ; lowers bacterial load	Established for gas gangrene, radiation necrosis, diabetic foot. Hypospadias data anecdotal ; cost and number of sessions weigh against it
Subcutaneous tissue expansion	Expander under shaft skin, filled via a subinguinal port every 1–2 weeks	Creates non-hair-bearing, androgen-receptor-bearing skin. Largest series: enough skin in 95% , but complications 41% (28% from the expander)

GRAFT MATERIALS FOR SALVAGE URETHROPLASTY

Graft	Advantages	Problems	Status
Extragenital full-thickness skin	Available	Shrinkage, scarring, stricture in single-stage use	Limited
Bladder mucosa	Replaces a scarce plate	Contracts in air → meatal narrowing; prolapse (“cauliflower” meatus) up to 38% ; meatal and junctional stenosis	Largely replaced by buccal mucosa

Graft	Advantages	Problems	Status
Buccal mucosa (cheek)	Adapted to wet and dry, neovascularises, hairless, minimal contraction , low donor morbidity, keeps its histology ; does not contract with growth on a good bed	Donor: mouth tightness, numbness, motor deficit, scar, salivary duct obstruction; more glans dehiscence than lip	Graft of choice
Lip mucosa	Thinner → better in the glans ; fewer complications	Donor morbidity equal to cheek	Preferred within the glans

- **Staged buccal repair beats single-stage** tubularisation (better graft take). A **dorsal inlay** is better than an onlay or complete buccal tube in redo cases with poor blood supply (Fig. 56.31).

Buccal graft technique: harvest **10% larger than the bed** → leave the donor site to heal by secondary intention (closure adds pain, little benefit) → combine sources for long defects → **fenestrate with an 18- or 21-gauge needle** (less haematoma) and keep in saline → incise the recipient bed ventrally from the urethra through the glans to the planned meatus, **preserving dartos** (the blood supply) → suture along its axis without narrowing it → **quilting sutures** against haematoma → wait **≥6 months (up to 12)** before urethroplasty, checking take and contraction → **topical steroid** for contraction (from 1 month) → family “**graft massage**” from about **2 weeks** (vitamin E oil, silicone gel or petroleum jelly).

- **Results after completion:** **30–75% success**; **meatal stenosis is the commonest failure**, then stricture and breakdown.
- **Cecil-Culp** (Cecil 1946; Culp 1951): urethroplasty → **midline scrotal incision** → recess the repaired penis into the scrotal dartos bed, where it **parasitises the scrotal blood supply** → release at a second stage (Cecil: **6–8 weeks**).
- It fell from favour with single-stage repair and has returned for salvage. The concerns are **hair-bearing scrotal skin** and multiple procedures.
- **Modification:** delay release to **9–12 months**. This transfers more vascularised layers, usually avoids using scrotal skin, and **stretches deficient shaft skin** for tension-free closure.

Figure legend, Fig. 56.44 (Weiss et al., 2018) — After failed two-stage and buccal repairs with two coronal fistulae: the fistulae and glans were closed, and the **ventral skin defect was covered by attaching the shaft to a midline scrotal incision**. About **1 year** later the scar had softened and the skin stretched; after release, the skin was closed in the midline with no recurrence at 3 months.

KEY POINTS

- **Lip mucosa within the glans:** thinner, fewer complications.
- **Buccal mucosa is the graft of choice** in complex reconstruction (properties as tabulated).
- **Cecil modification** vascularises skin after multiple repairs.

Concept — Reoperation is a problem of **blood supply, not technique**. Each step (staging, vascular layers, buccal grafts on dartos, scrotal parasitisation, tissue expansion, nitroglycerin, hyperbaric oxygen) brings vascularised, non-hair-bearing tissue to a bed that previous surgery has impoverished.

FUTURE

- Progress depends on **honest reporting, standardised pre-, intra- and postoperative assessment and definitions**, and **multi-institutional collaboration** to power studies of complication sources.

- **Coaching and teamwork**, as in athletics, can refine technique for rare and difficult cases, particularly proximal hypospadias.
- **Priorities**: lower complication rates in severe hypospadias, **follow-up into adulthood**, and long-term data to set expectations.
- **Open questions**: the ideal meatal position; what curvature is acceptable; how much surgery patients will accept. One patient left with a **coronal meatus but a straight penis and normal voiding is satisfied** (Fig. 56.45). Others are long-term psychological effects and how best to educate families.
- **Technology**: peri-operative **videography and crowd-sourced assessment**; **AI and machine learning** on recorded procedures; genomic, proteomic and microbiome risk profiling; a **central registry** linking anatomy, technique, outcomes and PROs; tissue banking; and perhaps **printed replacement tissue**.